COVID19 - Final Project

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2026-09-09

```
url_in <- "https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/csse_covi
file_names <- c("time_series_covid19_confirmed_US.csv",  "time_series_covid19_confirmed_global.csv", "ti
urls <- str_c(url_in,file_names)
```

```
us_cases <- read_csv(urls[1])
```

```
## Rows: 3342 Columns: 1154
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr    (6): iso2, iso3, Admin2, Province_State, Country_Region, Combined_Key
## dbl (1148): UID, code3, FIPS, Lat, Long_, 1/22/20, 1/23/20, 1/24/20, 1/25/20...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
global_cases <- read_csv(urls[2])
```

```
## Rows: 289 Columns: 1147
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr    (2): Province/State, Country/Region
## dbl (1145): Lat, Long, 1/22/20, 1/23/20, 1/24/20, 1/25/20, 1/26/20, 1/27/20,...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
us_deaths <- read_csv(urls[3])
```

```
## Rows: 3342 Columns: 1155
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr    (6): iso2, iso3, Admin2, Province_State, Country_Region, Combined_Key
## dbl (1149): UID, code3, FIPS, Lat, Long_, Population, 1/22/20, 1/23/20, 1/24...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
global_deaths <- read_csv(urls[4])
```

```
## Rows: 289 Columns: 1147
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr    (2): Province/State, Country/Region
## dbl (1145): Lat, Long, 1/22/20, 1/23/20, 1/24/20, 1/25/20, 1/26/20, 1/27/20,...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

COVID19 Report

These datasets contain daily case and death numbers for COVID-19 around the world between January 2020 and March 2023

```
# tidying global data for cases and deaths and combining into one df
# rotated df so dates are rows and changed format of dates and some column names
global_cases <- global_cases %>%
  pivot_longer(cols = -c('Province/State','Country/Region','Lat','Long'),names_to = "Date", values_to =
global_deaths <- global_deaths %>%
  pivot_longer(cols = -c('Province/State','Country/Region','Lat','Long'),names_to = "Date", values_to =
global <- global_cases %>% full_join(global_deaths) %>% rename(Country_Region = `Country/Region`, Provi
```

```
## Joining with `by = join_by(`Province/State`, `Country/Region`, Lat, Long,
## Date)`
```

```
# filter out entries with no cases
global <- global %>% filter(Cases>0)
#accessing population data to add into global
uid_lookup_url<-"https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/csse_covid_19_data/UI
uid <- read_csv(uid_lookup_url) %>% select(c(Province_State, Country_Region, Population))
```

```
## Rows: 4321 Columns: 12
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (7): iso2, iso3, FIPS, Admin2, Province_State, Country_Region, Combined_Key
## dbl (5): UID, code3, Lat, Long_, Population
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
global <- global %>% left_join(uid, by=c("Province_State", "Country_Region"))
```

Global Analysis

```
summary(global)
```

```
##  Combined_Key       Province_State     Country_Region          Lat
##  Length:306827      Length:306827      Length:306827      Min.   :-71.950
##  Class :character   Class :character   Class :character   1st Qu.:  5.152
##  Mode  :character   Mode  :character   Mode  :character   Median : 22.167
##                                                           Mean   : 20.527
##                                                           3rd Qu.: 41.113
##                                                           Max.   : 71.707
##                                                           NA's   :1910
##       Long              Date                Cases               Deaths
##  Min.   :-178.12   Min.   :2020-01-22   Min.   :        1   Min.   :      0
##  1st Qu.: -19.02   1st Qu.:2020-12-12   1st Qu.:     1316   1st Qu.:      7
##  Median :  21.01   Median :2021-09-16   Median :    20365   Median :    214
##  Mean   :  23.15   Mean   :2021-09-11   Mean   :  1032863   Mean   :  14405
##  3rd Qu.:  88.09   3rd Qu.:2022-06-15   3rd Qu.:   271281   3rd Qu.:   3665
##  Max.   : 178.06   Max.   :2023-03-09   Max.   :103802702   Max.   :1123836
##  NA's   :1910
##    Population
##  Min.   :6.700e+01
##  1st Qu.:7.866e+05
##  Median :6.948e+06
##  Mean   :2.890e+07
##  3rd Qu.:2.914e+07
##  Max.   :1.380e+09
##  NA's   :6729
```

Global cases and deaths by location

```
global_by_region <- global %>% group_by(Combined_Key)%>% mutate(New_Cases = Cases - lag(Cases), New_Dea
global_by_region
```

```
## # A tibble: 306,827 x 9
## # Groups:   Combined_Key [288]
##    Combined_Key   Lat  Long Date       Cases Deaths Population New_Cases
##    <chr>        <dbl> <dbl> <date>     <dbl>  <dbl>      <dbl>     <dbl>
##  1 Afghanistan   33.9  67.7 2020-02-24     5      0   38928341        NA
##  2 Afghanistan   33.9  67.7 2020-02-25     5      0   38928341         0
##  3 Afghanistan   33.9  67.7 2020-02-26     5      0   38928341         0
##  4 Afghanistan   33.9  67.7 2020-02-27     5      0   38928341         0
##  5 Afghanistan   33.9  67.7 2020-02-28     5      0   38928341         0
##  6 Afghanistan   33.9  67.7 2020-02-29     5      0   38928341         0
##  7 Afghanistan   33.9  67.7 2020-03-01     5      0   38928341         0
##  8 Afghanistan   33.9  67.7 2020-03-02     5      0   38928341         0
##  9 Afghanistan   33.9  67.7 2020-03-03     5      0   38928341         0
## 10 Afghanistan   33.9  67.7 2020-03-04     5      0   38928341         0
## # i 306,817 more rows
## # i 1 more variable: New_Deaths <dbl>
```

```
# all rows in global with missing values
global_clean <- global_by_region %>% drop_na()
global_missing <- anti_join(global_by_region,global_clean)
```

```
## Joining with `by = join_by(Combined_Key, Lat, Long, Date, Cases, Deaths,
## Population, New_Cases, New_Deaths)`
```

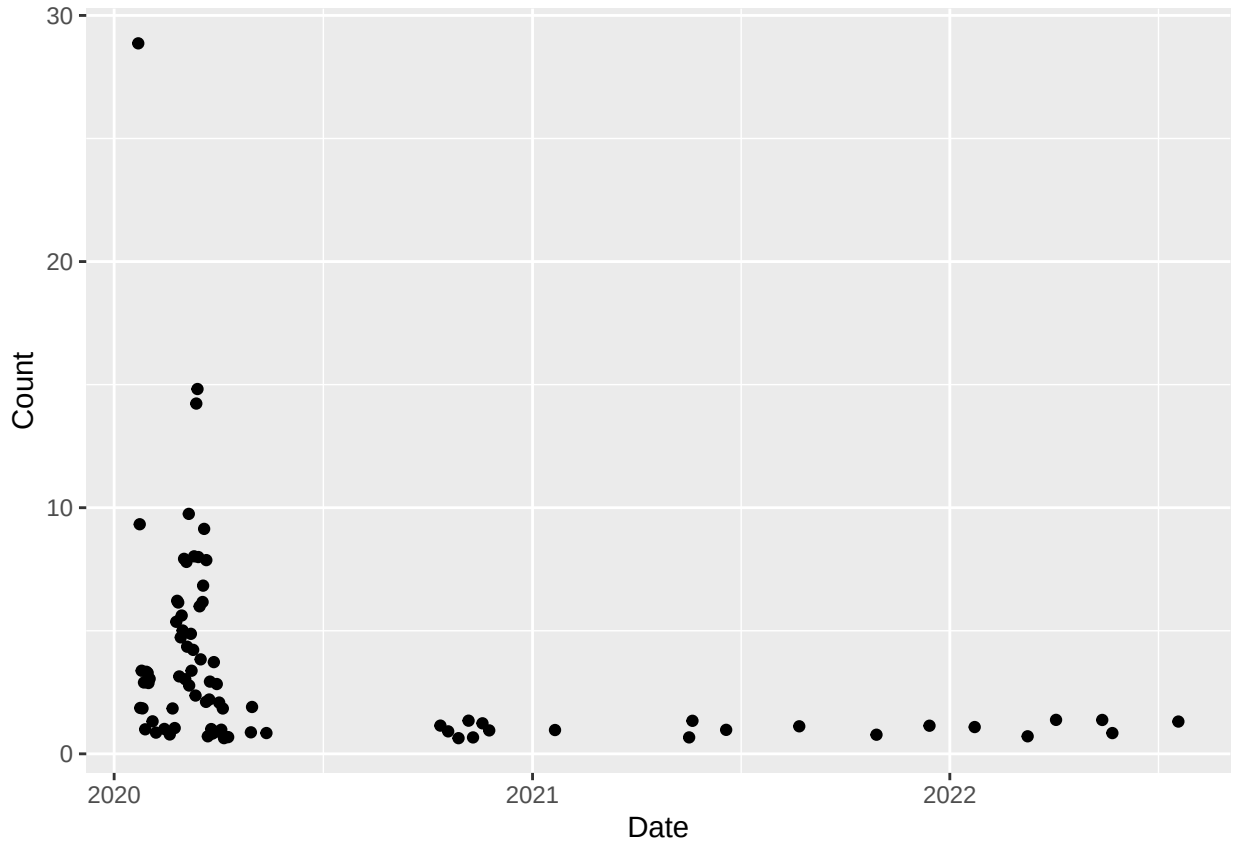
```r
# new cases = NA indicates first occurrence of that location in the dataset
global_first_cases <- global_missing %>% select(c(Combined_Key,Date,Cases,New_Cases,Population))
global_first_cases <- global_first_cases %>% filter(is.na(New_Cases))
global_first_cases
```

```
## # A tibble: 288 x 5
## # Groups:   Combined_Key [288]
##    Combined_Key                              Date       Cases New_Cases Population
##    <chr>                                     <date>     <dbl>     <dbl>      <dbl>
##  1 Afghanistan                               2020-02-24     5        NA   38928341
##  2 Albania                                   2020-03-09     2        NA    2877800
##  3 Algeria                                   2020-02-25     1        NA   43851043
##  4 Andorra                                   2020-03-02     1        NA      77265
##  5 Angola                                    2020-03-20     1        NA   32866268
##  6 Antarctica                                2021-12-14    11        NA         NA
##  7 Antigua and Barbuda                       2020-03-13     1        NA      97928
##  8 Argentina                                 2020-03-03     1        NA   45195777
##  9 Armenia                                   2020-03-01     1        NA    2963234
## 10 Australian Capital Territory, Australia 2020-03-13     1        NA     428100
## # i 278 more rows
```

When looking at new cases, the first occurrence of a particular location in the dataset shows new_cases as NA, so by filtering all the records where new_cases is NA, we get an idea of when each location reported their first cases of COVID-19.

First case reporting in each location

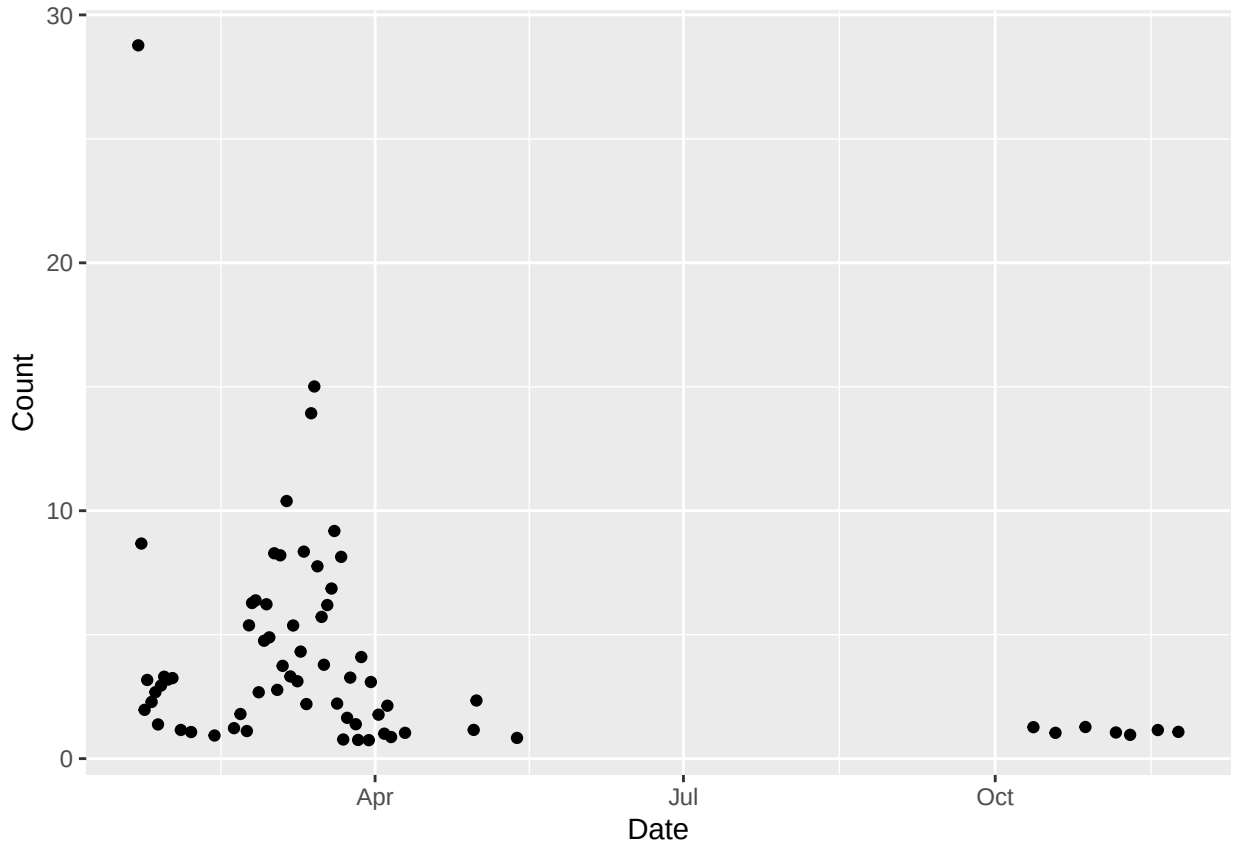
```r
first_cases <- global_first_cases %>% group_by(Date) %>% summarise(Count = n())
ggplot(first_cases,aes(Date,Count))+geom_jitter()
```



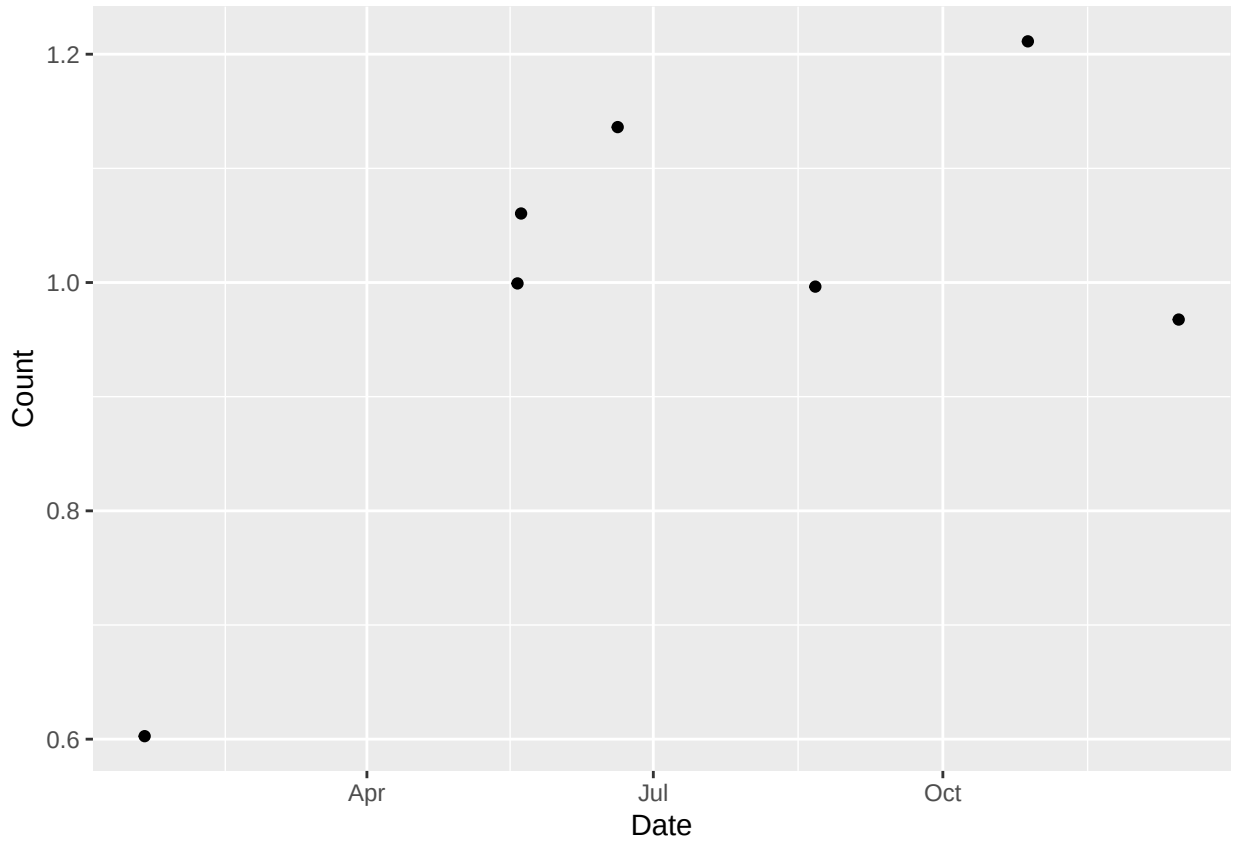
Unsurprisingly, the majority of locations reported their first cases in the first half of 2020.

Let's separate this out by year

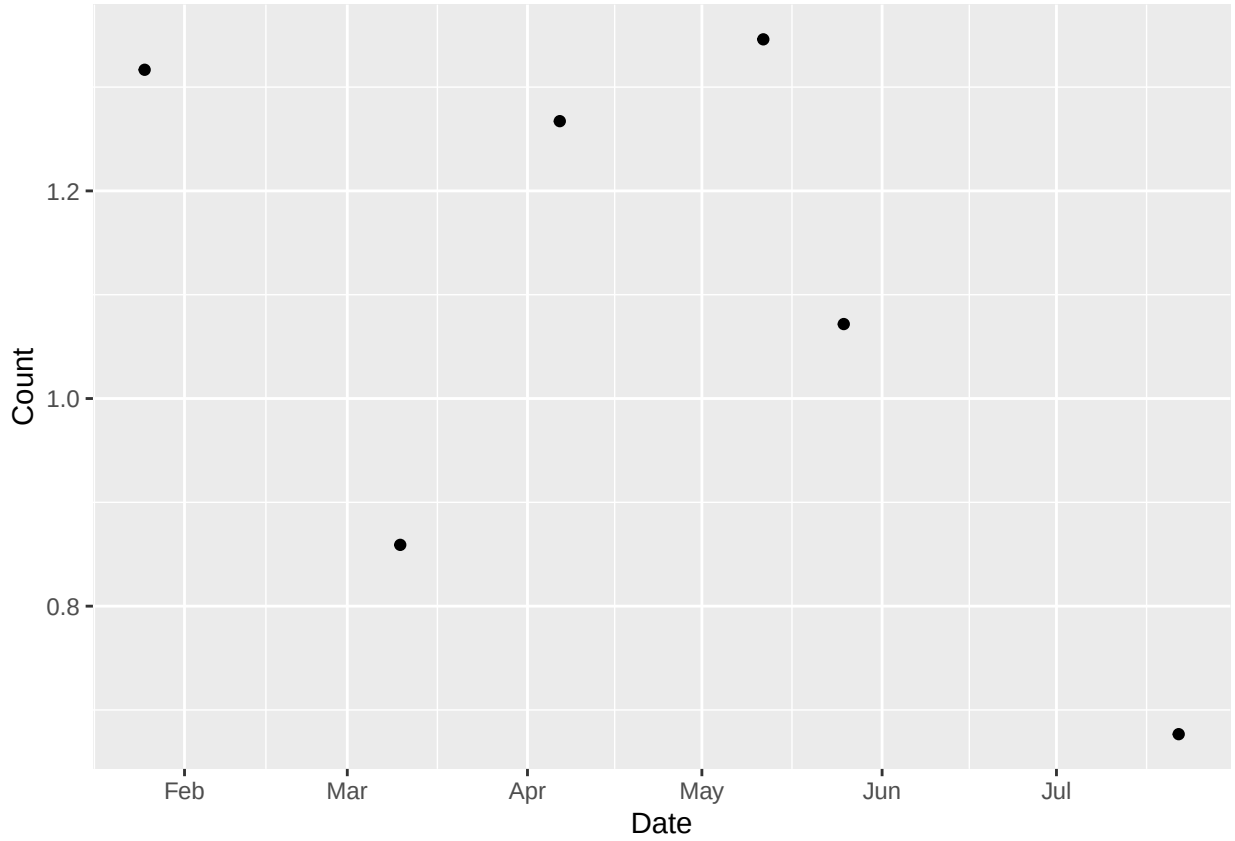
```
first_cases_20 <- global_first_cases %>% group_by(Date) %>% subset(year(Date) == 2020) %>% summarise(Cou
ggplot(first_cases_20,aes(Date,Count))+geom_jitter()
```




```
first_cases_21 <- global_first_cases %>% group_by(Date) %>% subset(year(Date) == 2021) %>% summarise(Cou
ggplot(first_cases_21,aes(Date,Count))+geom_jitter()
```




```
first_cases_22 <- global_first_cases %>% group_by(Date) %>% subset(year(Date) == 2022) %>% summarise(Co
ggplot(first_cases_22,aes(Date,Count))+geom_jitter()
```



The majority of "first" reported global cases in 2020 occurred between January and April, with a few locations starting their reporting in the fall. In 2021, first reportings were more spread out throughout the year. And in 2022, the final locations to report first cases were mostly in the first half of the year.

```
print(global_first_cases %>% subset(date(Date)==min(Date)),n=29)
```

```
## # A tibble: 29 x 5
## # Groups:   Combined_Key [29]
##    Combined_Key     Date       Cases New_Cases Population
##    <chr>            <date>     <dbl>     <dbl>      <dbl>
##  1 Anhui, China     2020-01-22     1        NA   61027171
##  2 Beijing, China   2020-01-22    14        NA   21893095
##  3 Chongqing, China 2020-01-22     6        NA   32054159
##  4 Fujian, China    2020-01-22     1        NA   41540086
##  5 Guangdong, China 2020-01-22    26        NA  126012510
##  6 Guangxi, China   2020-01-22     2        NA   50126804
##  7 Guizhou, China   2020-01-22     1        NA   38562148
##  8 Hainan, China    2020-01-22     4        NA   10081232
##  9 Hebei, China     2020-01-22     1        NA   74610235
## 10 Henan, China     2020-01-22     5        NA   99365519
## 11 Hubei, China     2020-01-22   444        NA   57752557
## 12 Hunan, China     2020-01-22     4        NA   66444864
## 13 Jiangsu, China   2020-01-22     1        NA   84748016
## 14 Jiangxi, China   2020-01-22     2        NA   45188635
## 15 Liaoning, China  2020-01-22     2        NA   42591407
## 16 Macau, China     2020-01-22     1        NA     649342
```

```
## 17 Ningxia, China   2020-01-22     1        NA    7202654
## 18 Shandong, China  2020-01-22     2        NA  101527453
## 19 Shanghai, China  2020-01-22     9        NA   24870895
## 20 Shanxi, China    2020-01-22     1        NA   34915616
## 21 Sichuan, China   2020-01-22     5        NA   83674866
## 22 Tianjin, China   2020-01-22     4        NA   13866009
## 23 Yunnan, China    2020-01-22     1        NA   47209277
## 24 Zhejiang, China  2020-01-22    10        NA   64567588
## 25 Japan            2020-01-22     2        NA  126476458
## 26 Korea, South     2020-01-22     1        NA   51269183
## 27 Taiwan*          2020-01-22     1        NA   23816775
## 28 Thailand         2020-01-22     4        NA   69799978
## 29 US               2020-01-22     1        NA  329466283
```

```
global_first_cases %>% subset(year(Date) == 2022)
```

```
## # A tibble: 6 x 5
## # Groups:   Combined_Key [6]
##   Combined_Key                      Date       Cases New_Cases Population
##   <chr>                             <date>     <dbl>     <dbl>      <dbl>
## 1 Korea, North                      2022-05-14     1        NA   25778815
## 2 Nauru                             2022-04-04     2        NA      10834
## 3 Niue, New Zealand                 2022-03-10     1        NA       1650
## 4 Tuvalu                            2022-05-23     3        NA      11792
## 5 Pitcairn Islands, United Kingdom  2022-07-20     4        NA         67
## 6 Winter Olympics 2022              2022-01-23    72        NA         NA
```

This is all the locations that had reported cases January 22,2020 and all the locations that reported their first cases in 2022.

Let's look at some outliers

```
global_by_region %>% subset(New_Deaths>10000)
```

```
## # A tibble: 3 x 9
## # Groups:   Combined_Key [2]
##   Combined_Key     Lat  Long Date         Cases Deaths Population New_Cases
##   <chr>          <dbl> <dbl> <date>       <dbl>  <dbl>      <dbl>     <dbl>
## 1 Chile          -35.7 -71.5 2022-03-21 3401145  55965   19116209      8167
## 2 Unknown, China  NA    NA   2023-01-15 1521816  59895         NA         0
## 3 Unknown, China  NA    NA   2023-01-21 1521816  72553         NA         0
## # i 1 more variable: New_Deaths <dbl>
```

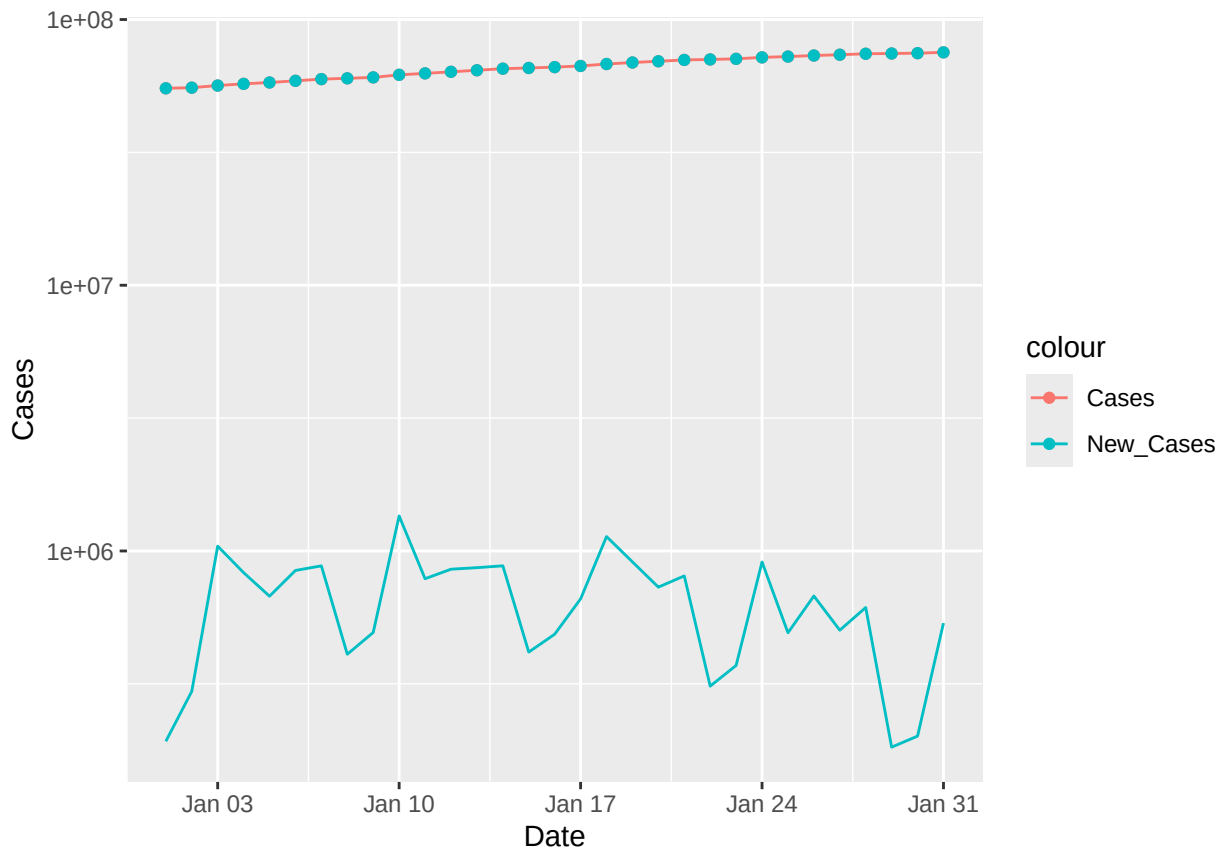
```
global_by_region %>% subset(New_Deaths<(-500))
```

```
## # A tibble: 2 x 9
## # Groups:   Combined_Key [2]
##   Combined_Key   Lat   Long Date         Cases Deaths Population New_Cases
##   <chr>        <dbl>  <dbl> <date>       <dbl>  <dbl>      <dbl>     <dbl>
## 1 Peru         -9.19 -75.0  2021-07-10 2078815 193230   32971846      4629
## 2 Spain        40.5   -3.75 2020-05-25  235400  26834   46754783      -372
## # i 1 more variable: New_Deaths <dbl>
```

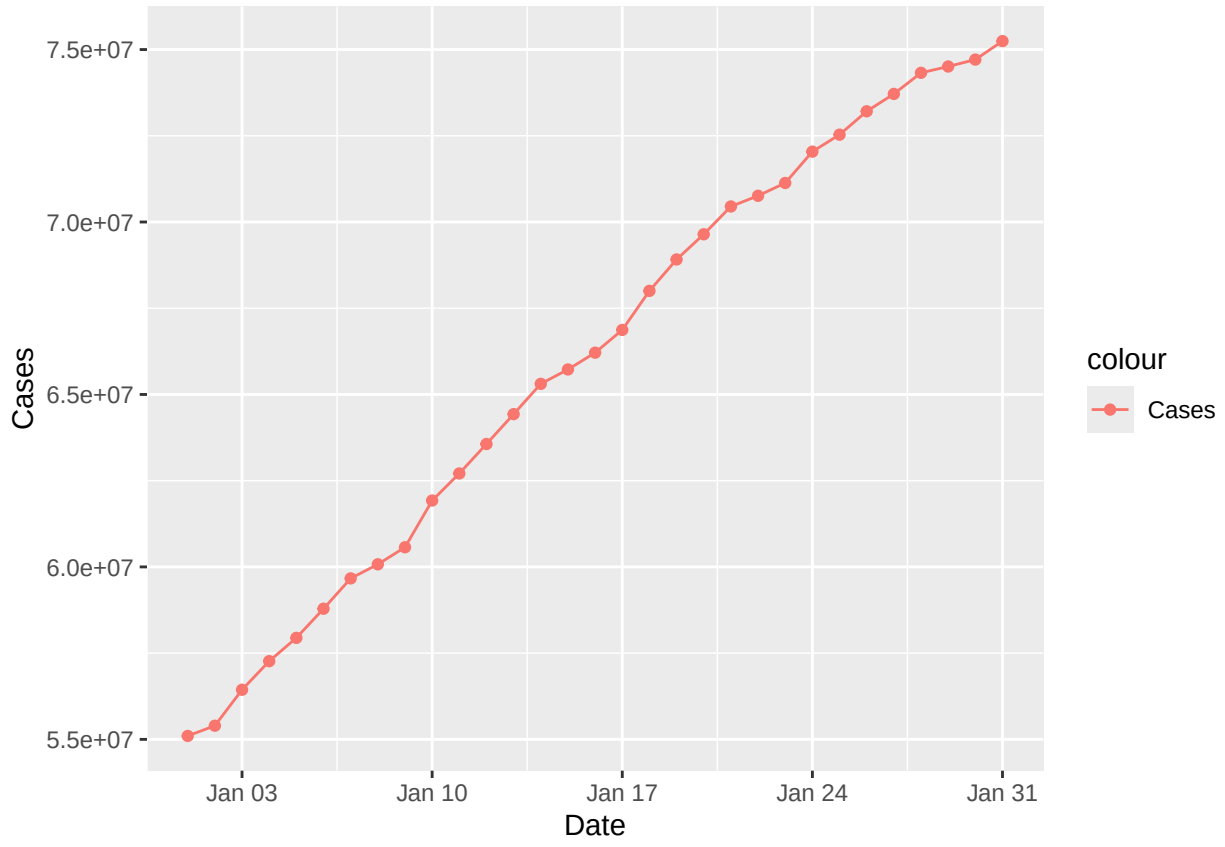
```
global_by_region %>% subset(New_Cases>800000)
```

```
## # A tibble: 14 x 9
## # Groups:   Combined_Key [3]
##    Combined_Key     Lat    Long Date          Cases Deaths Population New_Cases
##    <chr>          <dbl>   <dbl> <date>        <dbl>  <dbl>      <dbl>     <dbl>
##  1 Turkey          39.0   35.2  2020-12-10  1748567  15751   84339067    823225
##  2 US              40   -100    2022-01-03 56438983 828234  329466283   1042792
##  3 US              40   -100    2022-01-04 57267794 830557  329466283    828811
##  4 US              40   -100    2022-01-06 58788182 834879  329466283    844606
##  5 US              40   -100    2022-01-07 59667177 837319  329466283    878995
##  6 US              40   -100    2022-01-10 61923770 840861  329466283   1354505
##  7 US              40   -100    2022-01-12 63563852 846231  329466283    853867
##  8 US              40   -100    2022-01-13 64429141 848792  329466283    865289
##  9 US              40   -100    2022-01-14 65308292 851588  329466283    879151
## 10 US              40   -100    2022-01-18 68003787 857136  329466283   1132846
## 11 US              40   -100    2022-01-19 68913628 860621  329466283    909841
## 12 US              40   -100    2022-01-21 70450244 866932  329466283    805737
## 13 US              40   -100    2022-01-24 72039869 870881  329466283    908747
## 14 United Kingdom  55.4   -3.44 2022-01-31 17315893 184108   67886004    847371
## # i 1 more variable: New_Deaths <dbl>
```

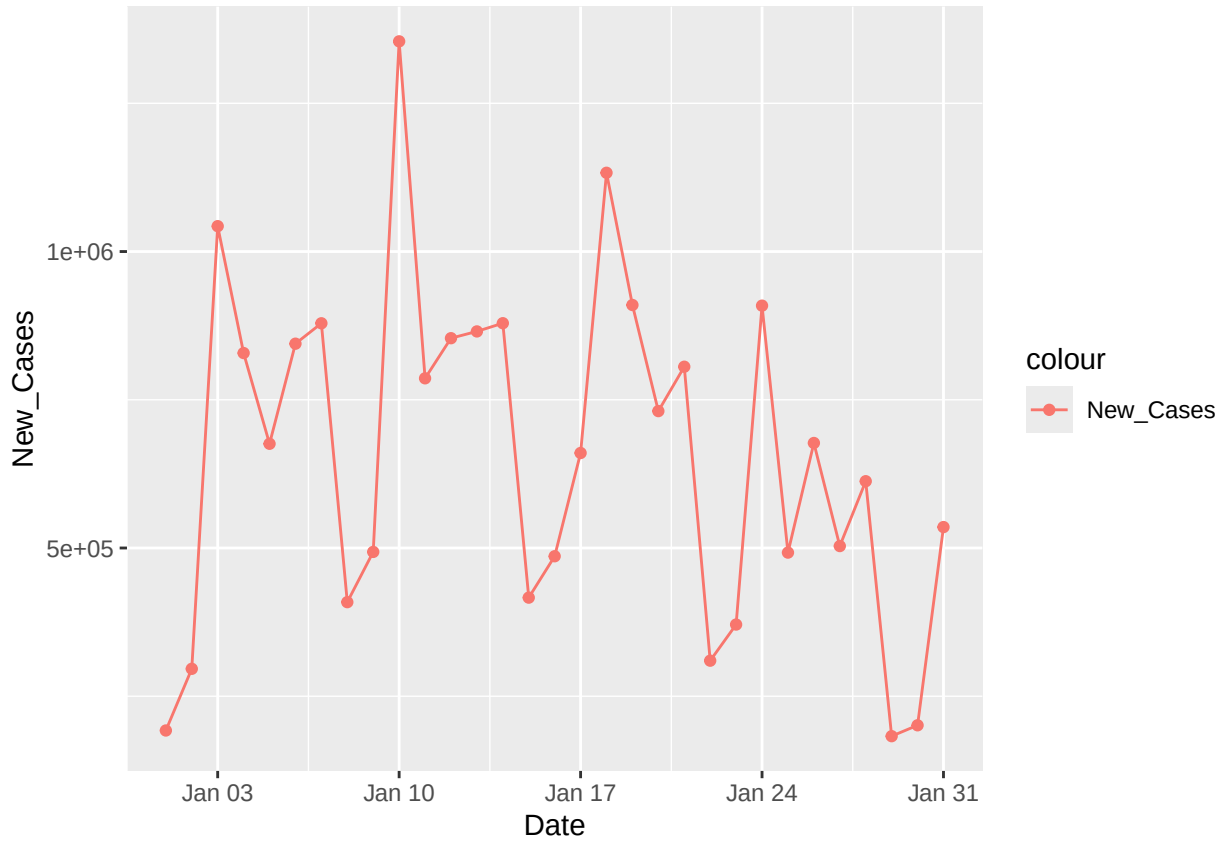
```
#looking at the month of January 2022 for US cases since the new cases were quite high for several date
globes <- global_by_region %>% subset(Combined_Key == 'US') %>% subset(month(Date) == 01 & year(Date) ==
ggplot(globes, aes(x=Date, y=Cases))+ geom_line(aes(color="Cases")) + geom_point(aes(color="Cases")) + g
```



```
ggplot(globes, aes(x=Date, y=Cases))+ geom_line(aes(color="Cases")) + geom_point(aes(color="Cases"))
```




```
ggplot(globes, aes(x=Date, y=New_Cases))+ geom_line(aes(color="New_Cases")) + geom_point(aes(color="New
```




```
#Looking at the new cases graph above, the largest increases occur roughly every 7 days with a somewhat

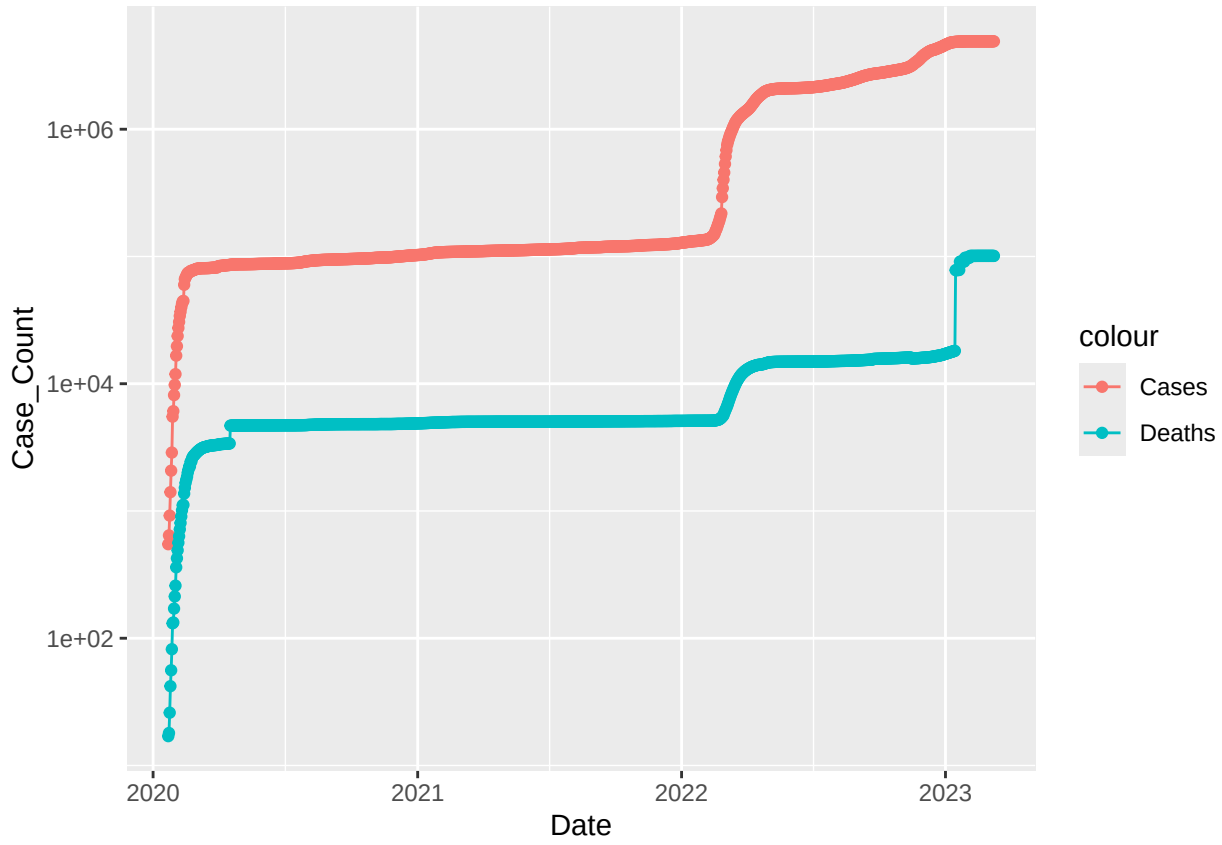
global_by_region %>% subset(New_Cases<(-50000))
```

```
## # A tibble: 5 x 9
## # Groups:   Combined_Key [5]
##   Combined_Key          Lat   Long Date        Cases Deaths Population New_Cases
##   <chr>               <dbl>  <dbl> <date>      <dbl>  <dbl>      <dbl>     <dbl>
## 1 New South Wales, ~ -33.9  151.   2022-01-31 1.09e6   1422    8118000    -53304
## 2 France              46.2    2.21 2021-05-20 5.51e6 107522   65249843   -349116
## 3 Kazakhstan          48.0   66.9  2021-07-24 5.82e5   8538   18776707    -59375
## 4 Peru                -9.19 -75.0  2022-12-25 4.38e6 218072   32971846    -65286
## 5 Spain               40.5   -3.75 2021-03-02 3.13e6  69801   46754783    -74347
## # i 1 more variable: New_Deaths <dbl>
```

The github page README recorded data modifications, retrospective reporting, back distributions, and update schedules. One of the records in the global set listed Spain as having -1918 deaths on May 25, 2020. This reduction in cases and deaths was due to data validation efforts. The new case value for France on 05/20/2021 is clearly an outlier. The note from the github for that record says it was a reduction of duplicate cases but the historical data had not been adjusted to match.

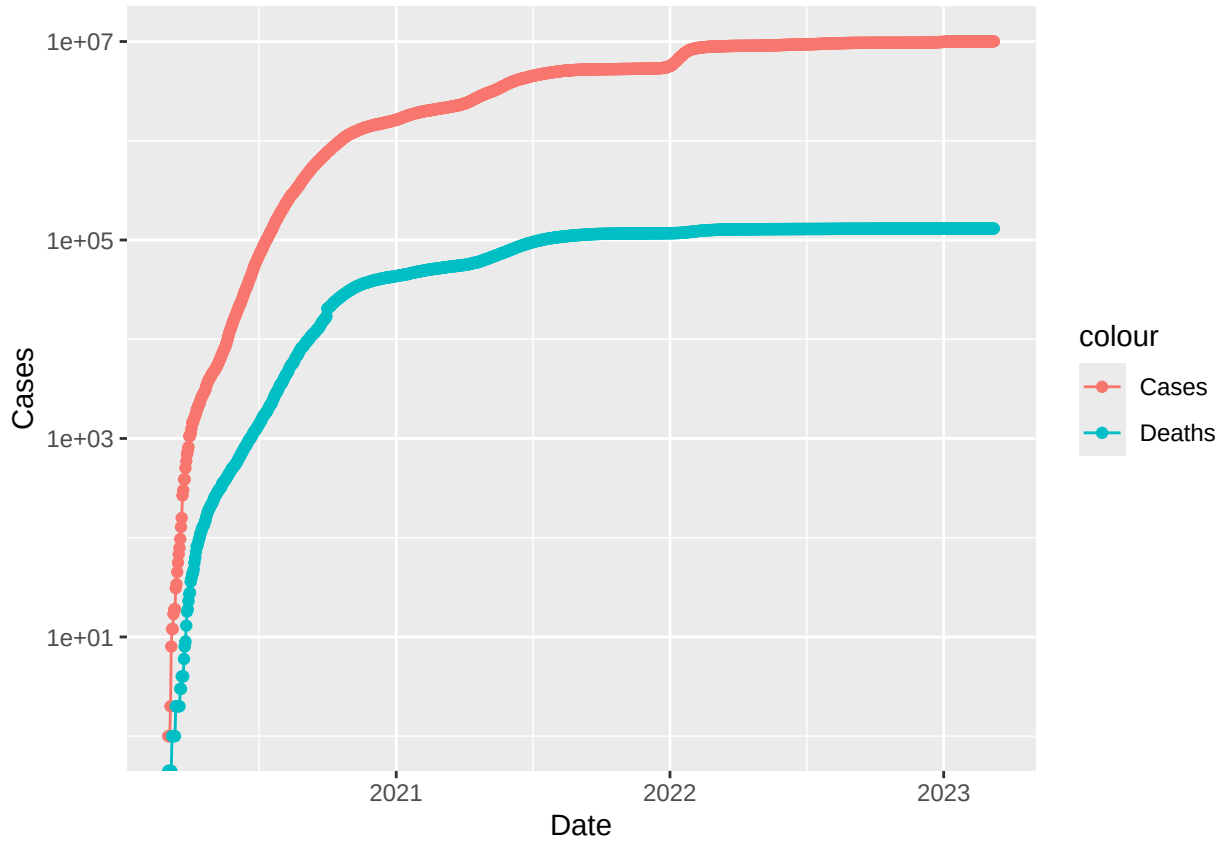
Now let's look at a few different regions

```
#China reporting
chi_reporting <- global_by_region %>% filter(str_detect(Combined_Key,'China')) %>% group_by(Date) %>% su
ggplot(chi_reporting, aes(x=Date, y=Case_Count))+ geom_line(aes(color="Cases")) + geom_point(aes(color=
```

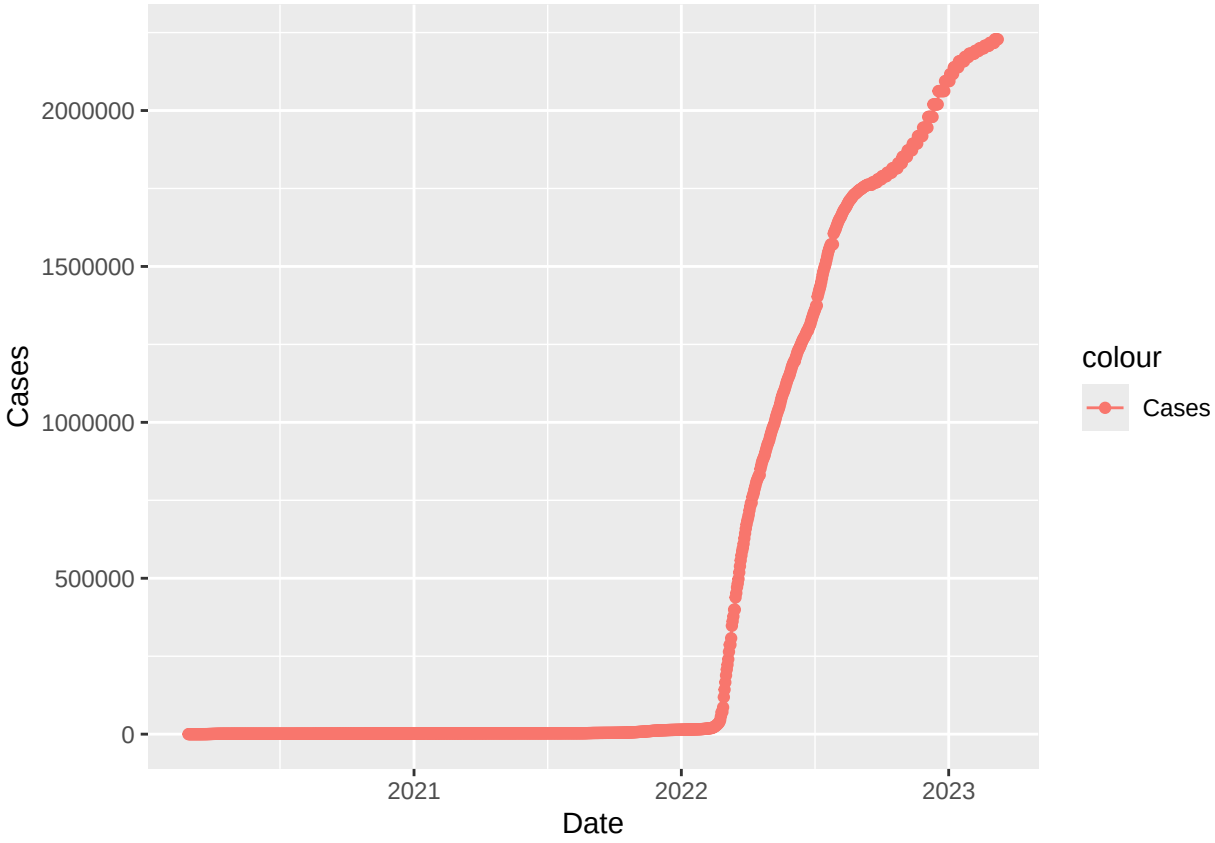



```
#Argentina reporting
arg_reporting <- global_by_region %>% subset(Combined_Key == 'Argentina')
ggplot(arg_reporting, aes(x=Date, y=Cases))+ geom_line(aes(color="Cases")) + geom_point(aes(color="Cases
```

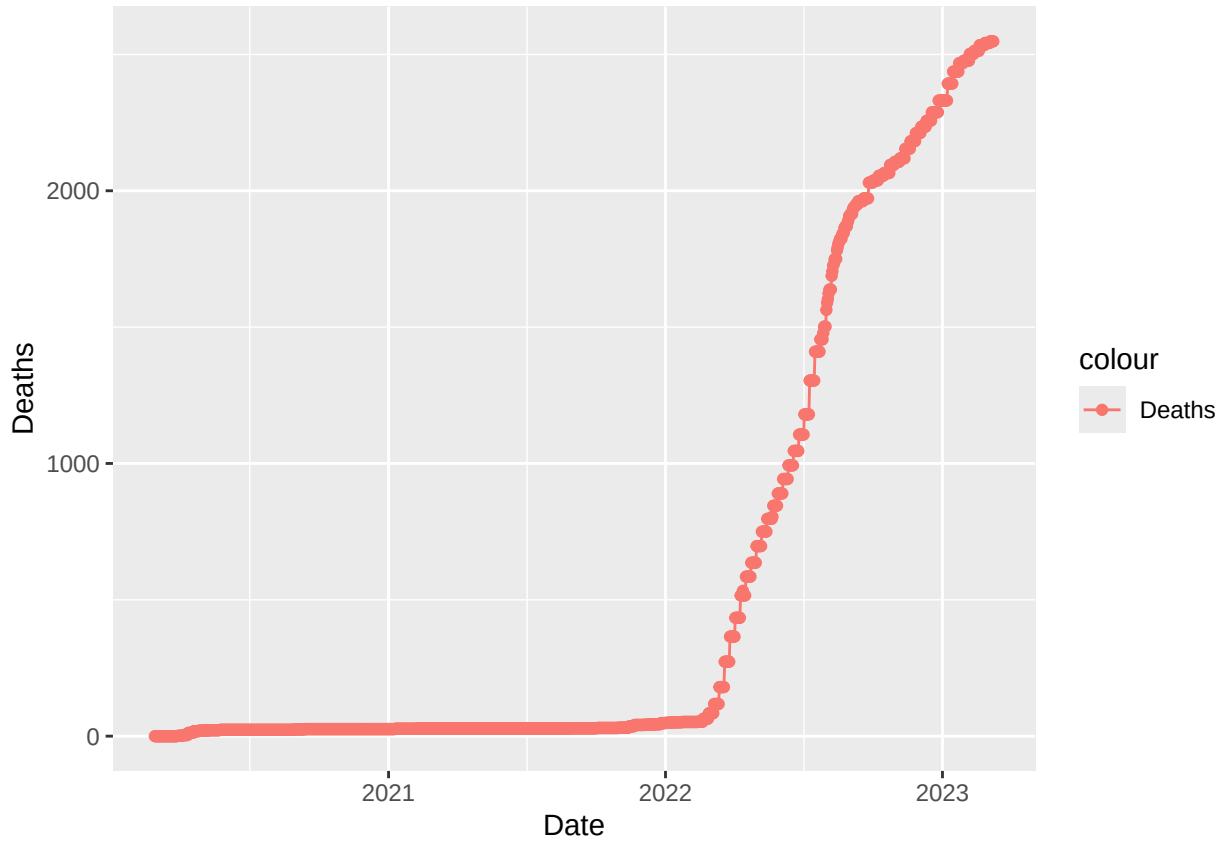
```
## Warning in scale_y_log10(): log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
```



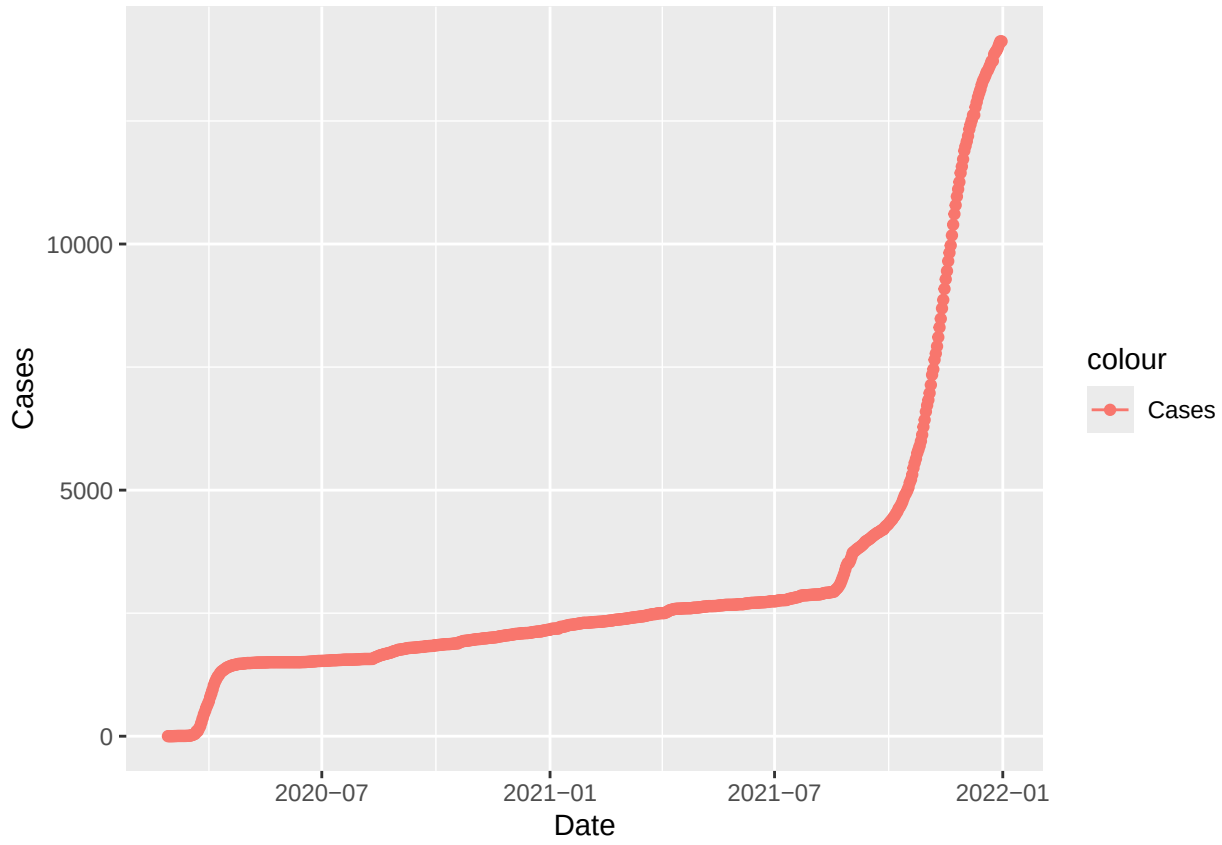

```
# New Zealand reporting. Not including log scale because it distorts the case/death increase
nz_reporting <- global_by_region %>% subset(Combined_Key == 'New Zealand')
ggplot(nz_reporting, aes(x=Date, y=Cases))+ geom_line(aes(color="Cases")) + geom_point(aes(color="Cases
```



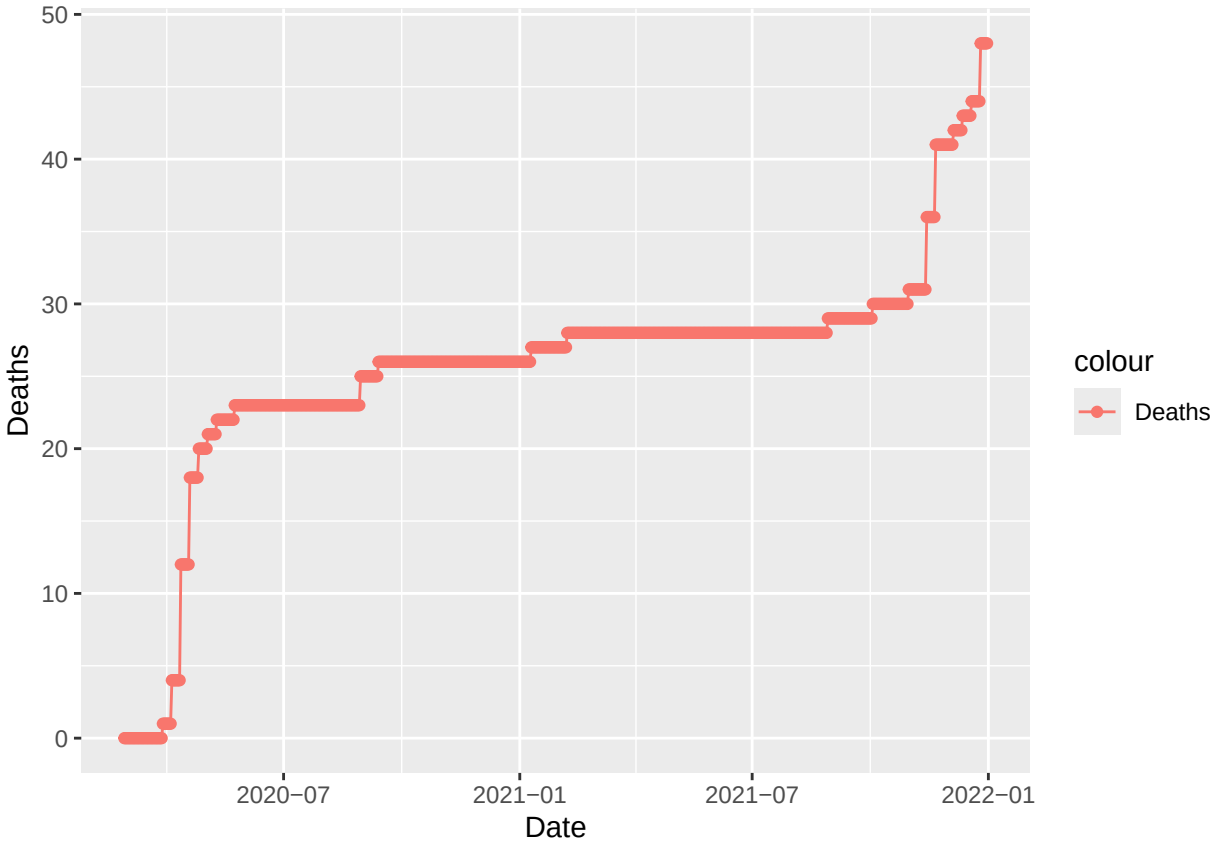

```
ggplot(nz_reporting, aes(x=Date, y=Deaths))+ geom_line(aes(color="Deaths")) + geom_point(aes(color="Dea
```




```
#subset
nz_reporting_sub <- global_by_region %>% subset(Combined_Key == 'New Zealand') %>% subset(year(Date)<20
ggplot(nz_reporting_sub, aes(x=Date, y=Cases))+ geom_line(aes(color="Cases")) + geom_point(aes(color="C
```




```
ggplot(nz_reporting_sub, aes(x=Date, y=Deaths))+ geom_line(aes(color="Deaths")) + geom_point(aes(color=
```



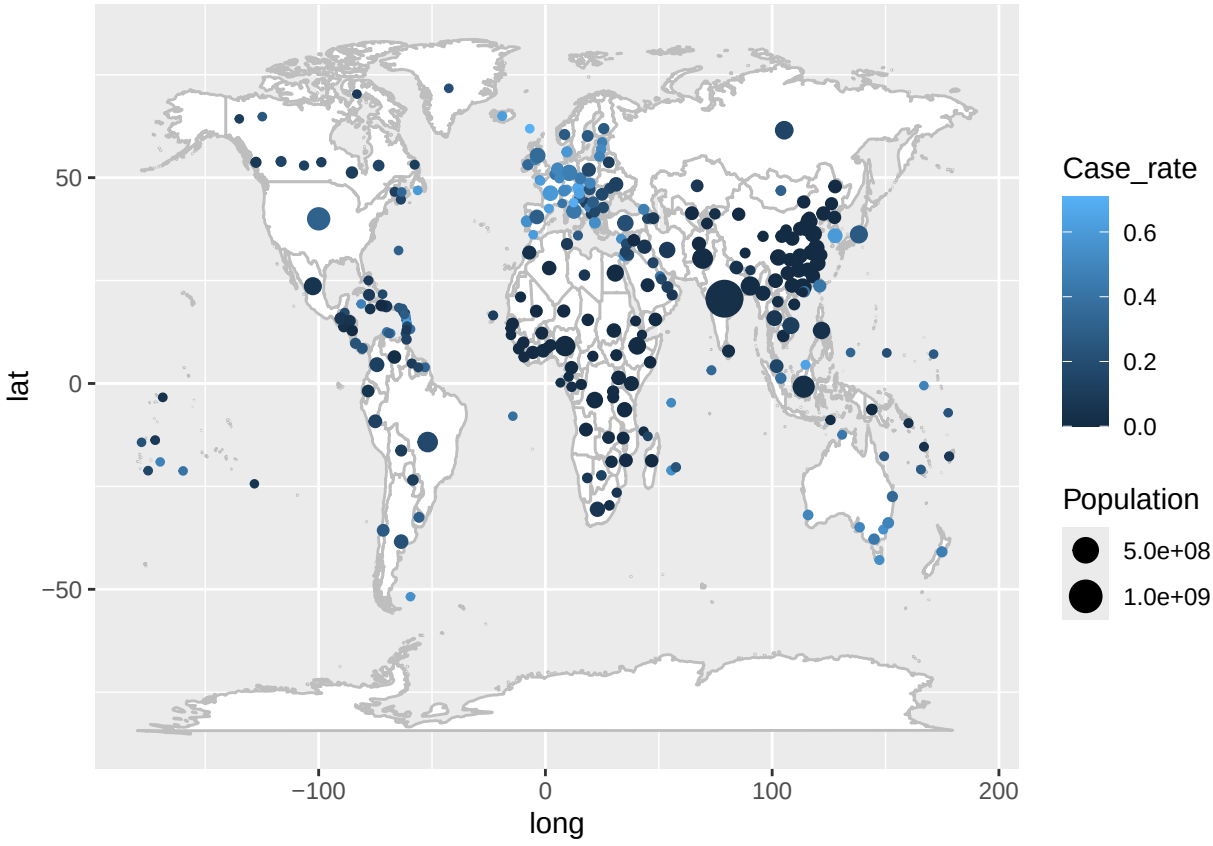
China reported the first cases of COVID and had a large initial outbreak. The government implemented strict testing and lockdown requirements in an effort to control the spread. Argentina had a long lockdown in Buenos Aires. New Zealand had a very early and very strict lockdown and didn't break 100 deaths until March 2022, at which point, reporting for deaths was weekly instead of daily.

Map - showing percentage of population who got and who died from COVID (note: not death rate)

```
world_coords <-map_data("world")
final_case <- global_by_region %>% subset(date(Date)==max(Date)) %>% mutate(Case_rate = round(Cases/Popu
ggplot()+geom_map(data=world_coords,map=world_coords,aes(long,lat,map_id=region),color='grey',fill='whi
```

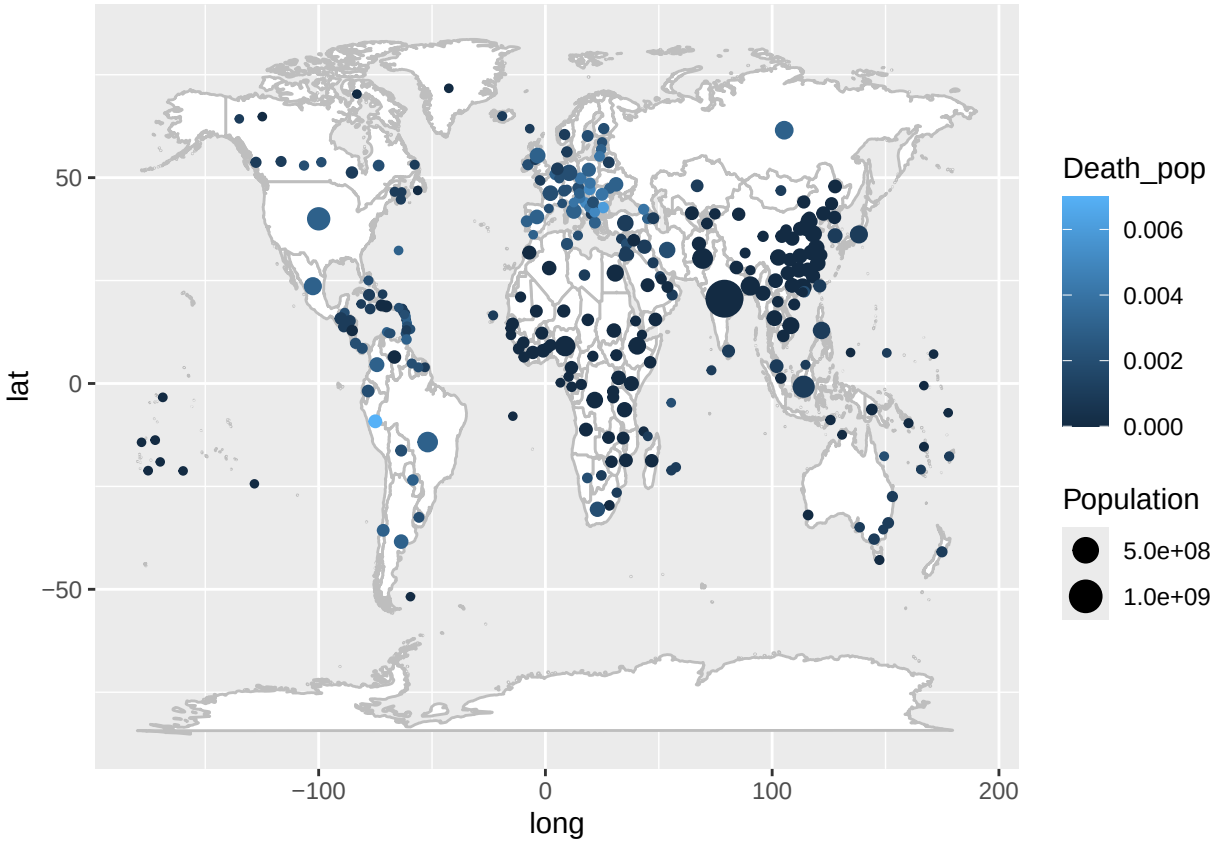
```
## Warning in geom_map(data = world_coords, map = world_coords, aes(long, lat, :
## Ignoring unknown aesthetics: x and y
```

```
## Warning: Removed 8 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
ggplot()+geom_map(data=world_coords,map=world_coords,aes(long,lat,map_id=region),color='grey',fill='whi
```

```
## Warning in geom_map(data = world_coords, map = world_coords, aes(long, lat, : Ignoring unknown aesth
## Removed 8 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Based on the map, most of Europe had high relative case rates. Even though number wise, places like India and China have significantly higher case counts, they also have significantly higher populations. Case rate was calculated by dividing number of cases by the population. For deaths, I did not calculate the relative death rate (death count/case count) but rather the overall percentage of the population who died (death count/population). This was because of issues with case reporting in some locations. For example, North Korea reported 1 case and 6 deaths.

Looking at the linear relationship between deaths and cases

```
per_mil <- global_by_region %>% group_by(Combined_Key) %>% summarize(deaths=max(Deaths), cases=max(Cases
```

```
## Warning: Returning more (or less) than 1 row per `summarise()` group was deprecated in
## dplyr 1.1.0.
## i Please use `reframe()` instead.
## i When switching from `summarise()` to `reframe()`, remember that `reframe()`
##   always returns an ungrouped data frame and adjust accordingly.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## `summarise()` has grouped output by 'Combined_Key'. You can override using the
## `.groups` argument.
```

```
model <- lm(deaths_per_mil~cases_per_mil, data=per_mil)
summary(model)
```

```
##
```

```
## Call:
## lm(formula = deaths_per_mil ~ cases_per_mil, data = per_mil)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2745.2  -289.0  -257.7    71.2  5788.1
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.890e+02  1.623e+00   178.0   <2e-16 ***
## cases_per_mil 4.271e-03  1.107e-05   385.9   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 775.4 on 300096 degrees of freedom
## Multiple R-squared:  0.3316, Adjusted R-squared:  0.3316
## F-statistic: 1.489e+05 on 1 and 300096 DF,  p-value: < 2.2e-16
```

Bias

The first date of reporting for this dataset began January 22, 2020. There had already been a growing number of cases worldwide since late 2019 so it would be more accurate to say that these are the first reportings included in this dataset rather than the first known cases. The WHO declared the coronavirus outbreak as a public health emergency January 30, 2020 and a pandemic March 11, 2020. Since we all lived through it recently, it can be difficult to separate ourselves from the data. For my part I tried to avoid getting into subjective or opinionated analysis on issues like if the lockdowns or travel restrictions were effective or not. But rather just look at the numbers. That being said, I did notice that some country reportings looked off. For instance, North Korea, as mentioned earlier, only reported 1 case and 6 deaths. Turkmenistan is the only country that did not report any cases. Both countries claimed to have very few/none because of immediate travel restrictions and such. However, given how quickly and widely it spread, it seems suspicious of there being no cases. Additionally reports were being made from governments or agencies which may have reasons to want to modify or conceal information. That could very well be in part to my own assumptions or the biases of the news and reporters. Looking through the github README, it became clear just how big of an undertaking this was. Hundreds of regions around the world reporting daily case and death numbers for over 2 years. It also explained some of the concerns and issues I had with some of the records during the analysis. I noticed there were differences and changes in reporting, backlogged tests, mistakes in data input, technical issues, duplicate counting, false positive/negatives, and more that contributed to some of the questionable data. Additionally, the way cases were counted changed since at first it was only lab confirmed cases vs probable or self reported cases/deaths. This dataset had some interesting things that would be missed if you drop NA immediately. For example, cases were reported from Antarctica, the Summer and Winter Olympics, a block of records for "Unknown China", and several repatriated travelers in Canada. There were also several cruise ships, which did report locations and some cases/deaths but don't have a population.